Forecasting Early-Stage Startup Growth:

Predicting Series B Attainment from Public Crunchbase Data

Final Project Report

Case Studies in Machine Learning

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Abstract

This project asks whether publicly available startup data can be used to forecast a concrete early-growth milestone: reaching Series B within 36 months of first observed funding. The data come from a Crunchbase-derived dataset distributed through Kaggle. After applying an observation-window rule and restricting the sample to firms whose first funding occurred on or before January 1, 2015 (so that the full 36-month horizon could actually be observed), I was left with 8,249 startups for analysis. Of these, 36.96% reached the Series B milestone within the window. The target itself is a practical proxy rather than a gold standard, because the dataset does not record the exact date of the Series B event; the label is built from round indicators together with the distance between first and last recorded funding. I compare four supervised models—logistic regression, random forest, histogram-based gradient boosting, and a multilayer perceptron—under two feature specifications and two evaluation schemes. The strict specification uses information that is plausibly available near the first funding event, while the expanded specification adds broader funding-history variables. Evaluation is done twice: once with a random train-test split and once with a forward-looking temporal split, so I can check whether the conclusions still hold when the model has to generalize to later cohorts. Across every experimental setting, tree-based ensembles beat the linear and neural baselines. Histogram-based gradient boosting is the strongest model family overall: the temporal expanded setup gives the best PR-AUC, and the temporal strict setup gives the best F1. The results also show that feature design matters at least as much as model choice. Letting the models see broader funding-history variables gives a real lift in predictive performance, but it also pushes the task away from a pure early-stage forecast. Taken together, these findings suggest that milestone prediction from public data is feasible, but that researchers should be upfront about how the label is constructed, how time is handled in evaluation, and where the line sits between predictive power and forecasting realism.

1. Introduction and Research Background

Forecasting startup outcomes is appealing for obvious reasons. Founders want to know whether their company is gaining the traction required for follow-on financing. Investors want a disciplined way to screen opportunities under extreme uncertainty. Policymakers and ecosystem builders want to understand which firms are more likely to translate early financing into sustained growth. Yet the prediction problem is difficult because startup trajectories are noisy, heterogeneous, and path-dependent. A company can look promising on paper and still fail to execute, while another can start slowly and later accelerate because of market timing, team changes, or a single strategic customer. That combination of economic importance and statistical difficulty has made startup prediction a natural use case for machine learning (Ross et al., 2021; Kim et al., 2023; Razaghzadeh Bidgoli et al., 2024; Corea, 2021; Zhang et al., 2019; Schade et al., 2023).

Much of the recent literature has focused on broad labels such as successful exit, acquisition, IPO, or generic “startup success.” Those labels are useful, but they can also be hard to interpret. Acquisition, for example, may represent a highly successful exit in one case and a distressed sale in another. Similarly, a simple success-versus-failure classification often mixes firms at very different stages of maturity. For this project, I chose a narrower and more operational milestone: whether a startup reaches Series B within 36 months of first recorded funding. This target is easier to connect to venture decision-making because Series B generally signals that a company has moved beyond seed-stage experimentation and has persuaded investors that it is ready to scale (Te et al., 2023; Mars Discovery District, 2026).

The project is also motivated by an important methodological issue in this literature: leakage. Several startup-prediction studies report strong performance, but later work has pointed out that some models benefit from variables that are direct or near-direct consequences of later success, which makes the task artificially easy (Żbikowski & Antosiuk, 2021; Potanin et al., 2023). That critique matters for a class project because a model that predicts well for the wrong reason is less useful than a slightly weaker model with a cleaner design. In response, I explicitly compare a strict feature set built around early information with an expanded feature set that includes broader funding-history variables. The comparison makes it possible to ask not only which model predicts best, but also how much of the predictive gain comes from relaxing the information boundary.

A second design choice concerns evaluation. Startup data are inherently temporal. Investors observe firms founded in one period and then make judgments that play out in later periods. Because of that, a purely random train-test split can overstate real-world performance if it allows the model to learn patterns that are specific to a historical period but not stable in future cohorts. The recent startup-prediction literature increasingly recommends time-aware validation for precisely this reason (Roberts et al., 2017; Wei et al., 2025). Accordingly, I evaluate the models twice: once with a random split and once with a forward-looking temporal split. This allows the paper to assess whether the ranking of models survives a more realistic evaluation design.

The study is organized around three research questions. First, can a public Crunchbase-derived dataset support useful prediction of Series B attainment within 36 months? Second, does model performance change meaningfully when the evaluation moves from random shuffling to a temporal split? Third, how much does predictive accuracy depend on feature scope, especially the inclusion of broader funding-history variables that are informative but less purely early-stage? To answer these questions, I train logistic regression as an interpretable baseline, random forest and histogram-based gradient boosting as tree-ensemble models, and a multilayer perceptron as a simple neural baseline. I use threshold tuning rather than a default 0.50 cutoff so that the models are assessed on a fairer decision basis in an imbalanced setting (He & Garcia, 2009; Saito & Rehmsmeier, 2015).

The contribution of this paper is modest but concrete. It does not claim to solve startup selection. Instead, it shows how a careful case-study design—clear milestone definition, explicit discussion of label limitations, separate feature specifications, threshold tuning, and time-aware evaluation—changes the interpretation of a startup-prediction project. In that sense, the project is aligned with the spirit of the course: the goal is not simply to maximize a leaderboard metric, but to use machine learning in a way that answers a meaningful applied question and makes its assumptions visible.

2. Materials and Data Sources

The data used in this project come from a publicly distributed Crunchbase-derived startup dataset hosted on Kaggle (Kaggle, n.d.-a, n.d.-b). The broader Crunchbase ecosystem is widely used in both academic research and industry because it contains structured information on company profiles, funding rounds, industries, locations, and exits (Crunchbase, 2026a, 2026b, 2026c). Public Kaggle mirrors are imperfect and often lag behind commercial Crunchbase products, but they remain useful for reproducible classroom research because they are accessible, structured, and widely reused in prior startup-prediction studies (Ross et al., 2021; Kim et al., 2023; Crunchbase News, 2023).

The raw file contained 54,294 startup records. Turning that into a clean prediction sample took two filters. First, every firm in the analysis sample needed to have non-missing first-funding and last-funding dates so that the time between rounds could actually be measured. Second, each firm needed a full 36-month window over which the milestone could either happen or not happen. I implemented this by requiring first_funding_at to fall on or before January 1, 2015, which gives each firm enough downstream history to be scored fairly against the 36-month target. Positives (firms that reached Series B within 36 months of first funding) were kept regardless, and negatives were kept only if their observed window from first to last funding was at least 36 months long, so the label is well-defined on both sides. After these filters, the usable sample was 8,249 startups. Within that sample, 36.96% of firms were labeled as having reached Series B within the 36-month window and 63.04% were not. Figure 1 shows the resulting class distribution. It is worth noting that this balance is not the same as the raw base rate of Series B in Crunchbase; it reflects the fact that the observation-window rule disproportionately removes young firms with short observed histories, which tend to be negatives. That is a deliberate tradeoff. The remaining sample is smaller and less imbalanced, but the label it supports is one that can actually be verified against a 36-month horizon.

The target label is an approximation rather than an event-history gold standard. The dataset does not include the exact date on which each startup entered Series B. Instead, the target was constructed from the presence of a Series B indicator together with the time difference between first and last recorded funding. In practical terms, a startup was treated as positive if the data indicated a Series B round and the observed funding timeline implied that this milestone was reached within 36 months of first funding. This approximation is a limitation and should be read as such. It makes the label useful for a case study, but not identical to a hand-validated venture financing timeline.

Missing data were concentrated in a few columns rather than spread evenly across the dataset. The highest missingness occurred in founding-date derivatives and the derived time-to-first-funding field, followed by website and category-related variables. Figure 2 summarizes the top missingness rates. For numeric variables, I imputed missing values conservatively with the median of the training data. For categorical variables, I used an explicit unknown category rather than dropping observations. This choice preserved sample size while making the missingness pattern visible to the models.

The market composition of the final sample was concentrated in a handful of sectors. Figure 3 shows that biotechnology and software were the largest groups, followed by enterprise software, health care, mobile, and several digital platform categories. This sector mix matters because venture financing norms vary by industry. Biotechnology, for example, often follows a different capital and timeline logic than software. The heavy presence of a few sectors therefore provides both signal and a potential source of bias. A model that performs well on this sample may be learning something real about venture milestones, but it may also partially reflect the historical composition of the dataset.

Before modeling, I also examined pairwise correlations among the main numeric variables. The strongest relationships were not surprising: founding year was strongly associated with the derived time-to-first-funding measure, and funding total was highly correlated with debt-financing-related fields. At the same time, no single numeric variable showed a strong bivariate correlation with the target itself. That pattern is important because it suggests the prediction task is driven less by one dominant linear signal and more by interactions and threshold effects among variables—precisely the setting in which tree ensembles often outperform linear models (Breiman, 2001; Friedman, 2001).

Figure 1. *Target class distribution in the final analysis sample after the 36-month observation filter.*

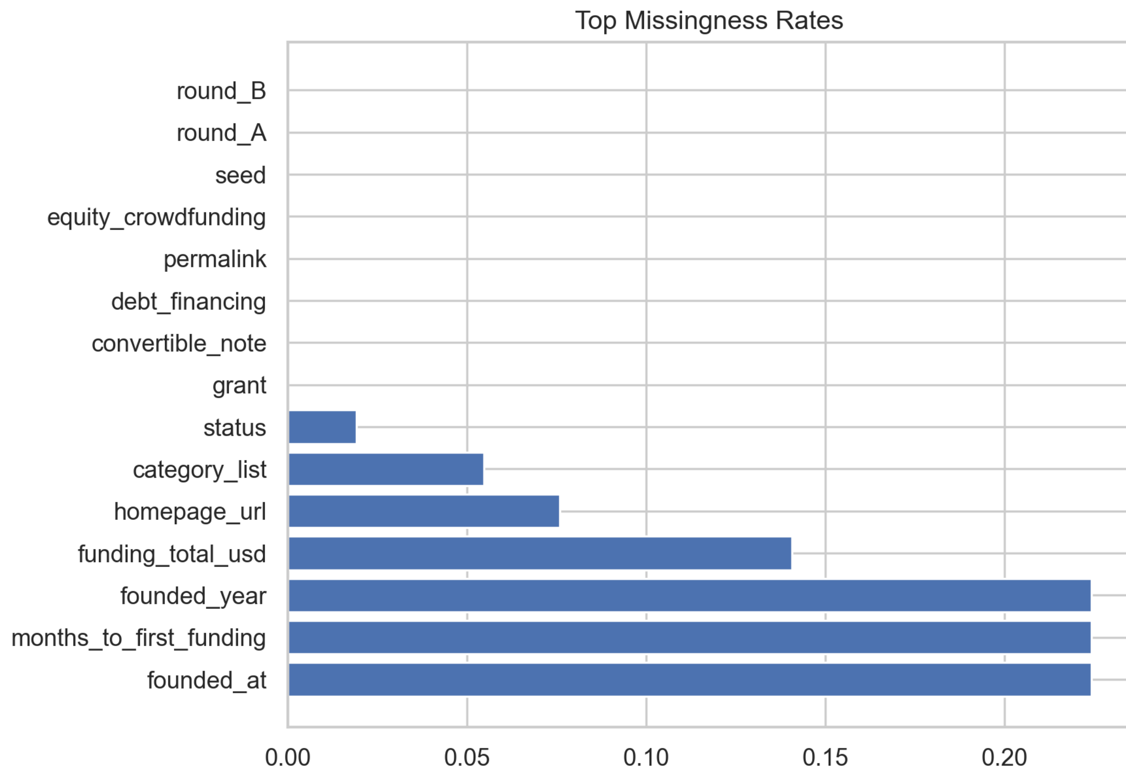

Figure 2. *Top missingness rates across the variables retained for modeling and reporting.*

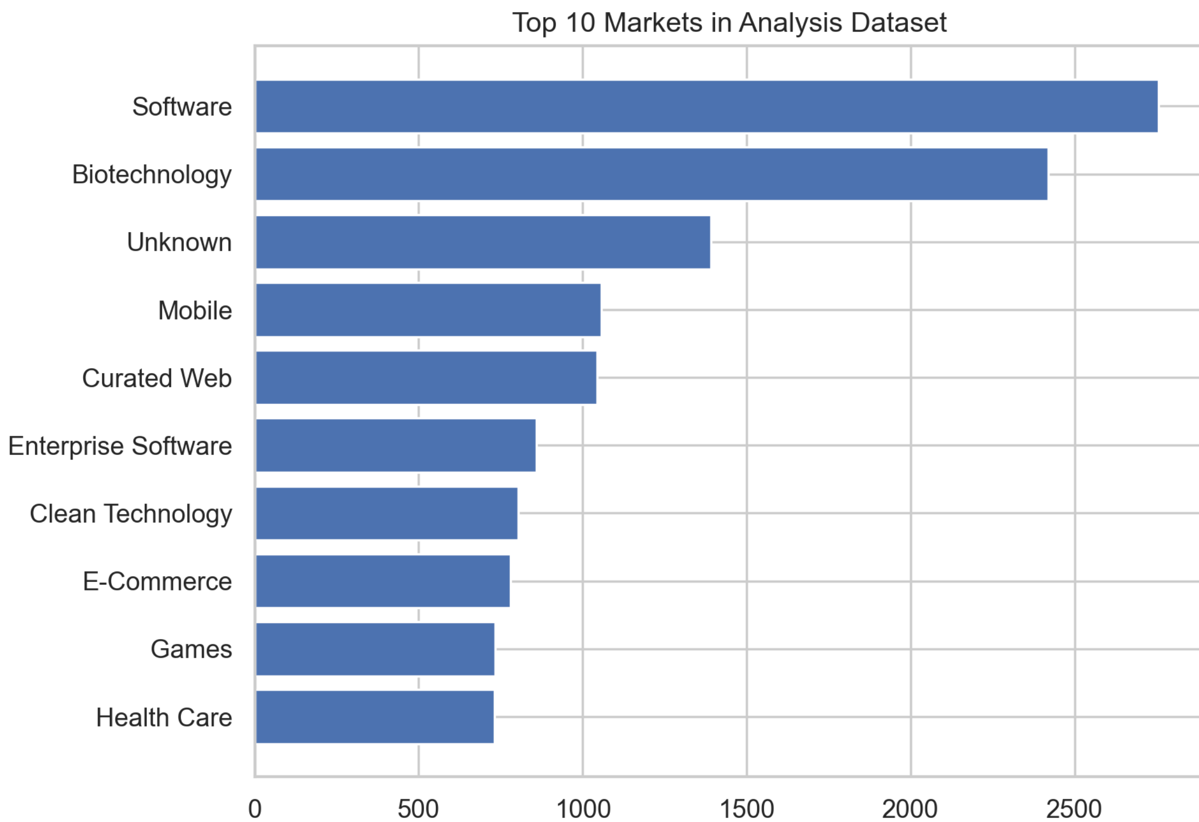

Figure 3. *Top market categories in the analysis sample. Biotechnology and software dominate the final cohort.*

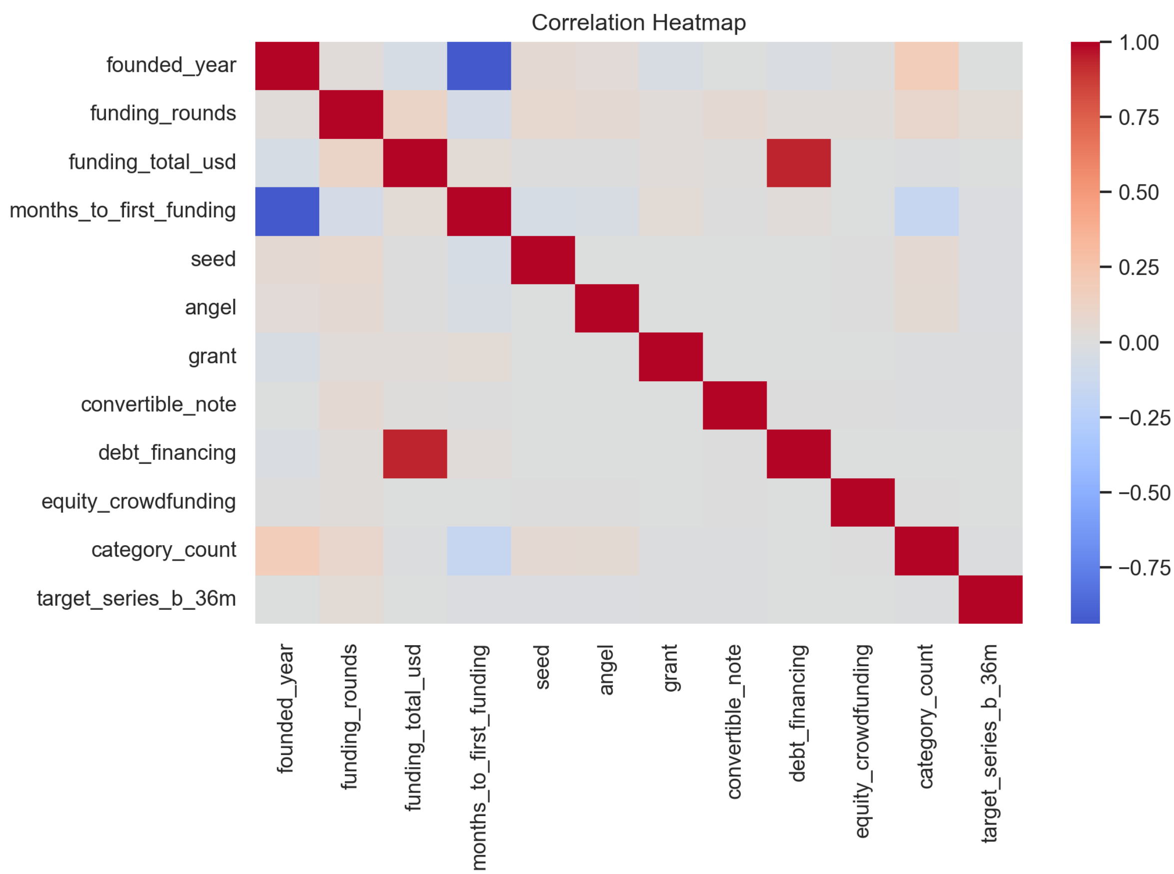

Figure 4. *Correlation heatmap for the main numeric variables and the milestone label.*

3. Research Design and Methods

The core prediction task is binary classification. Let y = 1 if a startup reaches Series B within 36 months of first recorded funding and y = 0 otherwise. I trained four supervised models that span a reasonable range of complexity for tabular data, all implemented in scikit-learn (scikit-learn developers, 2026a, 2026b, 2026c, 2026d). Logistic regression served as the baseline because it is familiar, easy to interpret, and provides a clear linear benchmark. Random forest was included because it handles nonlinear interactions and mixed feature types well while offering robust out-of-the-box performance on tabular problems (Breiman, 2001). Histogram-based gradient boosting was included because gradient boosting is frequently strong on structured business data and often outperforms simpler models when predictive patterns are nonlinear (Friedman, 2001). Finally, a multilayer perceptron was added as a shallow neural baseline. Although neural models can perform well in some startup-prediction settings, they are not automatically superior on modestly sized tabular datasets (Hastie et al., 2009; Kim et al., 2023).

The modeling pipeline used two feature specifications. The strict specification was designed to approximate a cleaner ex ante screening task. It used founding-year information, market and location variables, time-to-first-funding, category breadth, and early funding-type indicators such as seed, angel, grant, convertible note, debt financing, and equity crowdfunding. The expanded specification added broader funding-history variables, especially total funding and the number of funding rounds. This second specification was expected to improve prediction, but it also pushes the model closer to a snapshot description of a company that already carries some consequences of later development. The comparison between strict and expanded features is therefore substantive, not merely technical.

For evaluation, I used two train-test designs. In the random split, 80% of firms were assigned to training and 20% to testing through random sampling. In the temporal split, earlier firms were used for training and later firms were reserved for testing. The temporal design is more realistic because it better matches how a decision-maker would apply the model: learn from past cohorts and evaluate on future ones. Within the training data, I carved out a validation subset for threshold tuning. This matters because default thresholds are often poor choices in imbalanced classification, especially when the costs of false positives and false negatives are asymmetric (He & Garcia, 2009; Saito & Rehmsmeier, 2015).

I looked at six metrics: accuracy, precision, recall, F1, ROC-AUC, and PR-AUC. Accuracy on its own is a weak summary for this kind of task, because a model can post a respectable number by just leaning toward the larger class. Precision matters because investors care about the quality of what the model flags—they do not want most of the shortlist to be weak prospects. Recall matters because missing a promising firm is costly too. F1 was used as the decision metric during threshold tuning because it pulls precision and recall into a single operating score. ROC-AUC captures how well the classifier ranks firms across all possible thresholds, and PR-AUC gives a sharper read of the precision-recall tradeoff, which is the view that tends to matter more when the positive class is the minority (Saito & Rehmsmeier, 2015).

Interpretability was handled in two ways. First, I compared models under the strict and expanded feature sets, which makes the information boundary itself part of the interpretation. Second, I examined feature influence through model-based interpretation. A supplemental SHAP analysis was produced for histogram-based gradient boosting in the temporal setting, and a permutation-importance analysis was retained for random forest as a more conservative cross-check. In both cases, the purpose was not to assign causal meaning to the predictors, but to identify which variables the models relied on most heavily when ranking startups by their probability of hitting the Series B milestone (Lundberg & Lee, 2017; Molnar, 2022).

4. Results

Table 1 reports the main classification results. The overall ranking is stable and easy to summarize. Tree-based ensemble models dominate the baselines, the expanded feature specification improves performance relative to the strict specification, and histogram-based gradient boosting is the strongest model family overall. In other words, the answer to the first research question is yes: even with an approximate milestone label, publicly available startup data contain enough signal to support useful prediction of Series B attainment within 36 months.

Under the random split, the best model was histogram-based gradient boosting with the expanded feature set. It achieved an accuracy of 0.837, precision of 0.812, recall of 0.728, F1 of 0.768, ROC-AUC of 0.911, and PR-AUC of 0.884. Random forest was the next strongest model in the same specification, with an F1 of 0.728 and ROC-AUC of 0.860. The MLP and logistic regression also improved markedly once the expanded features were available, but they still trailed the leading tree models. In the strict specification, performance dropped for all four models, with the best random-split F1 falling to 0.669 for histogram-based gradient boosting. This pattern suggests that startup milestone prediction is possible even with cleaner early-stage information, but that the broader funding-history variables carry a sizable amount of signal.

The temporal split produced the most interesting findings. If the model had collapsed under forward-looking evaluation, that would have suggested the random split was overstating its usefulness. Instead, the ranking remained remarkably stable. Histogram-based gradient boosting with the expanded feature set again performed best, reaching an F1 of 0.767, ROC-AUC of 0.825, and PR-AUC of 0.943. Logistic regression in the expanded temporal setting was unexpectedly competitive on F1 at 0.764, but it did so with a very different balance of precision and recall. Random forest remained strong but no longer matched the best gradient-boosting result. In the strict temporal setup, histogram-based gradient boosting again led the table on F1 at 0.787, while random forest was a close second at 0.770.

The temporal results answer the second research question in an interesting way. Time-aware evaluation did not overturn the main conclusion that tree ensembles are best, but it did slightly reshape the relative performance of the models. In practical terms, this means the startup signal in the dataset was not purely an artifact of random shuffling. The best-performing models still generalized to later cohorts. That is an encouraging result for the project, because it suggests the model is learning patterns with at least some temporal stability rather than merely fitting cross-sectional noise.

The third research question was about feature scope, and the pattern here is clear. Expanded features beat strict features for every model family, and they do so across ROC-AUC, PR-AUC, and F1. This is not a surprising result on its own. Total funding and number of funding rounds are compact summaries of a company's financing path so far, and that path is obviously informative about whether a later financing milestone is likely. The more interesting question is what the comparison actually tells us. The expanded setup is the stronger predictor, but it is also less of a forecast in the strict sense, because it borrows from the trajectory it is trying to predict. The strict setup is weaker, but it is closer to the situation an investor is really in: judging a company using only what can be seen near the start of the financing path.

Figure 5, Figure 6, and Figure 7 visualize the model comparison across ROC-AUC, PR-AUC, and F1. Taken together, the charts reinforce two conclusions. First, histogram-based gradient boosting is the most reliable top performer. Second, the advantage of expanded features is not limited to a single metric. The same ordering appears whether the focus is ranking quality, precision-recall tradeoffs, or threshold-based operational performance. These charts are more useful than a single metric because they show that the preferred model is not an artifact of cherry-picking one evaluation criterion.

Interpretability adds another layer to the results. Figure 8 shows a SHAP beeswarm plot for the temporal expanded histogram-based gradient boosting model. Funding rounds, total funding, seed-related indicators, founding year, and time to first funding appear near the top of the ranking. The plot suggests that the model is not relying on one variable in isolation; rather, it is combining the scale, pace, and form of financing with sector and geography indicators. Figure 9 shows the same type of plot for the temporal strict model. Once the broader funding-history variables are removed, the importance ranking shifts toward early financing type, months to first funding, founding year, and certain sector indicators. This is exactly the substantive contrast the strict-versus-expanded design was meant to reveal.

Finally, Figure 10 provides a confusion matrix for the temporal strict histogram-based gradient boosting model after threshold tuning. The model still produces both false positives and false negatives, which is expected in a noisy startup setting, but it does not collapse into trivial majority-class predictions. That matters because a naive threshold choice can make a model look worse than it really is. In earlier exploratory runs, default thresholds caused some models to predict nearly all negatives. Threshold tuning corrected that issue and made the comparison much more meaningful.

Table 1. Main model results by split, feature specification, and algorithm

Split	Spec	Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random	Expanded	HistGradientBoosting	0.837	0.812	0.728	0.768	0.911	0.884
Random	Expanded	Random Forest	0.790	0.698	0.761	0.728	0.860	0.811
Random	Expanded	MLP	0.783	0.681	0.779	0.726	0.865	0.815
Random	Expanded	Logistic Regression	0.780	0.694	0.725	0.709	0.863	0.798
Random	Strict	HistGradientBoosting	0.717	0.590	0.770	0.669	0.814	0.747
Random	Strict	Random Forest	0.665	0.531	0.803	0.640	0.765	0.647
Random	Strict	Logistic Regression	0.660	0.531	0.697	0.603	0.731	0.611
Random	Strict	MLP	0.685	0.565	0.639	0.600	0.742	0.658
Temporal	Expanded	HistGradientBoosting	0.691	0.930	0.652	0.767	0.825	0.943
Temporal	Expanded	Logistic Regression	0.683	0.909	0.659	0.764	0.791	0.928
Temporal	Expanded	Random Forest	0.599	0.949	0.511	0.665	0.791	0.925
Temporal	Expanded	MLP	0.570	0.944	0.475	0.632	0.790	0.928
Temporal	Strict	HistGradientBoosting	0.696	0.865	0.722	0.787	0.714	0.882
Temporal	Strict	Random Forest	0.682	0.878	0.686	0.770	0.716	0.879
Temporal	Strict	Logistic Regression	0.560	0.846	0.531	0.652	0.636	0.843
Temporal	Strict	MLP	0.554	0.851	0.518	0.644	0.659	0.858

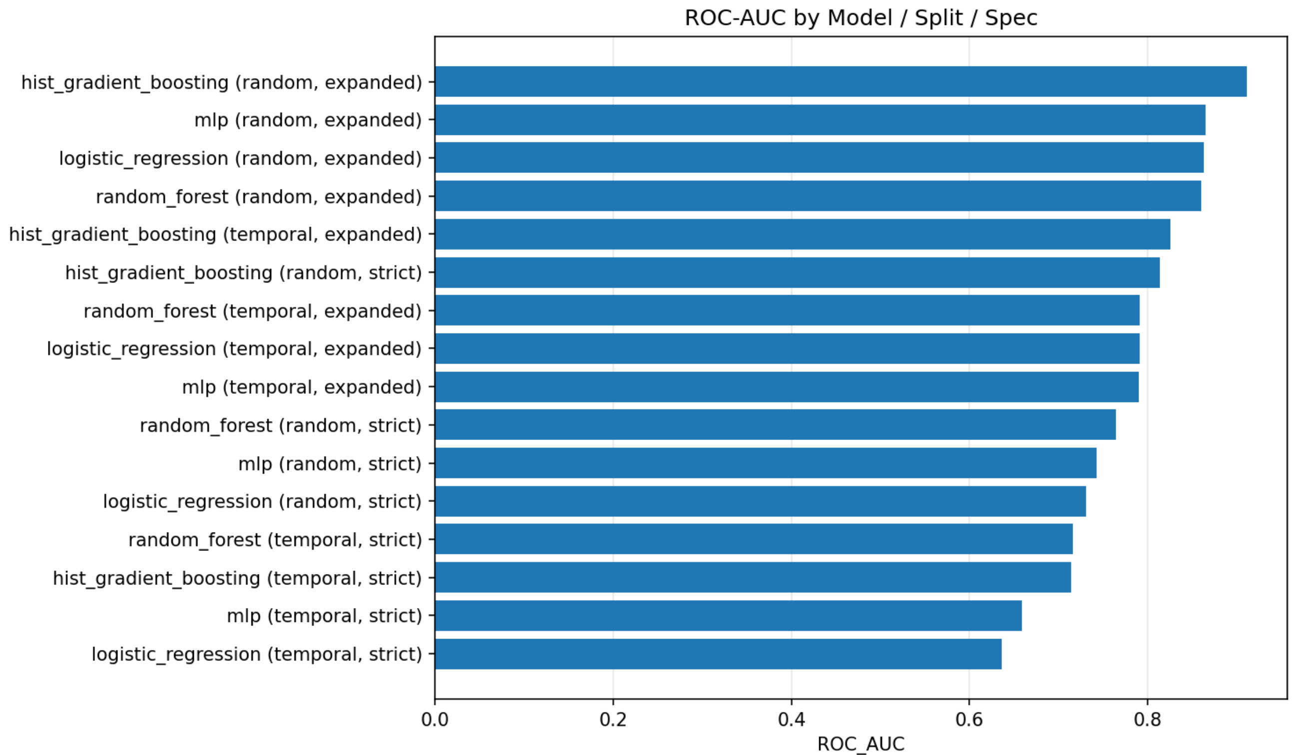

Figure 5. *ROC-AUC comparison across all models, splits, and feature specifications.*

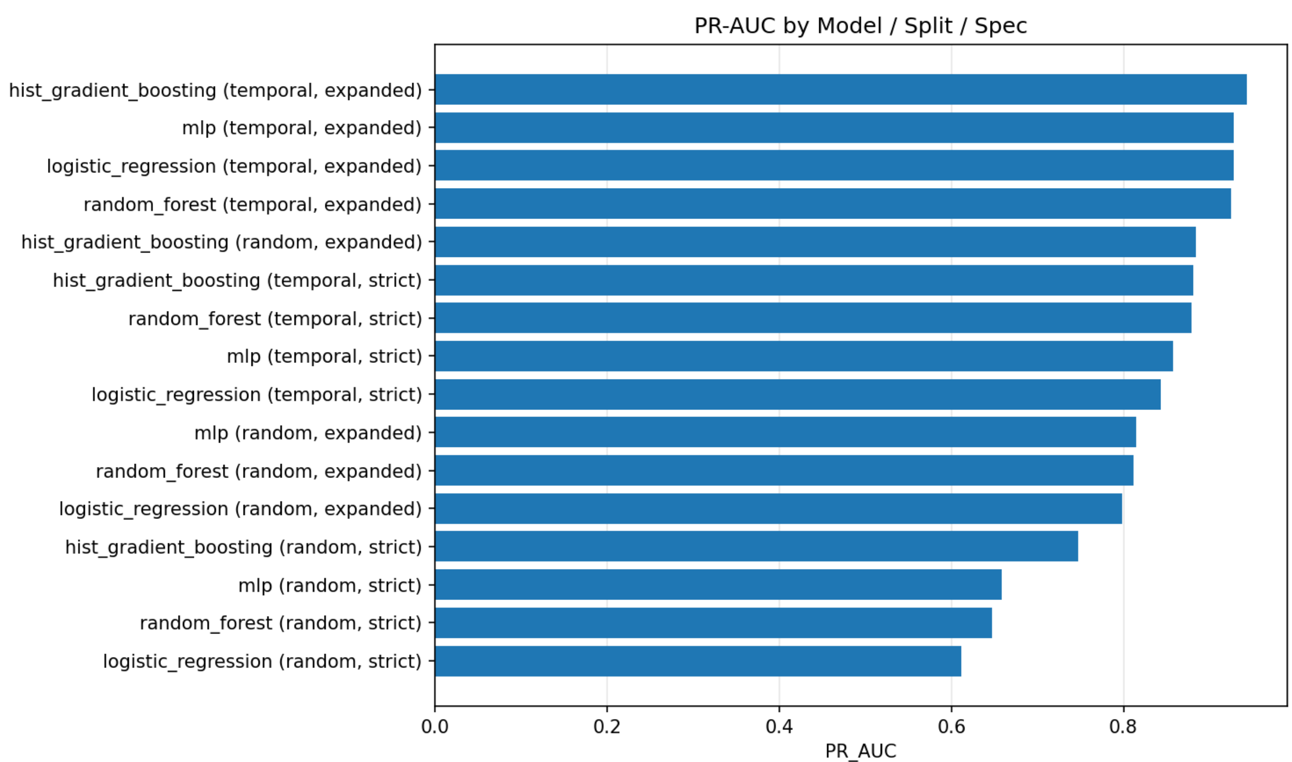

Figure 6. *PR-AUC comparison across all models, splits, and feature specifications.*

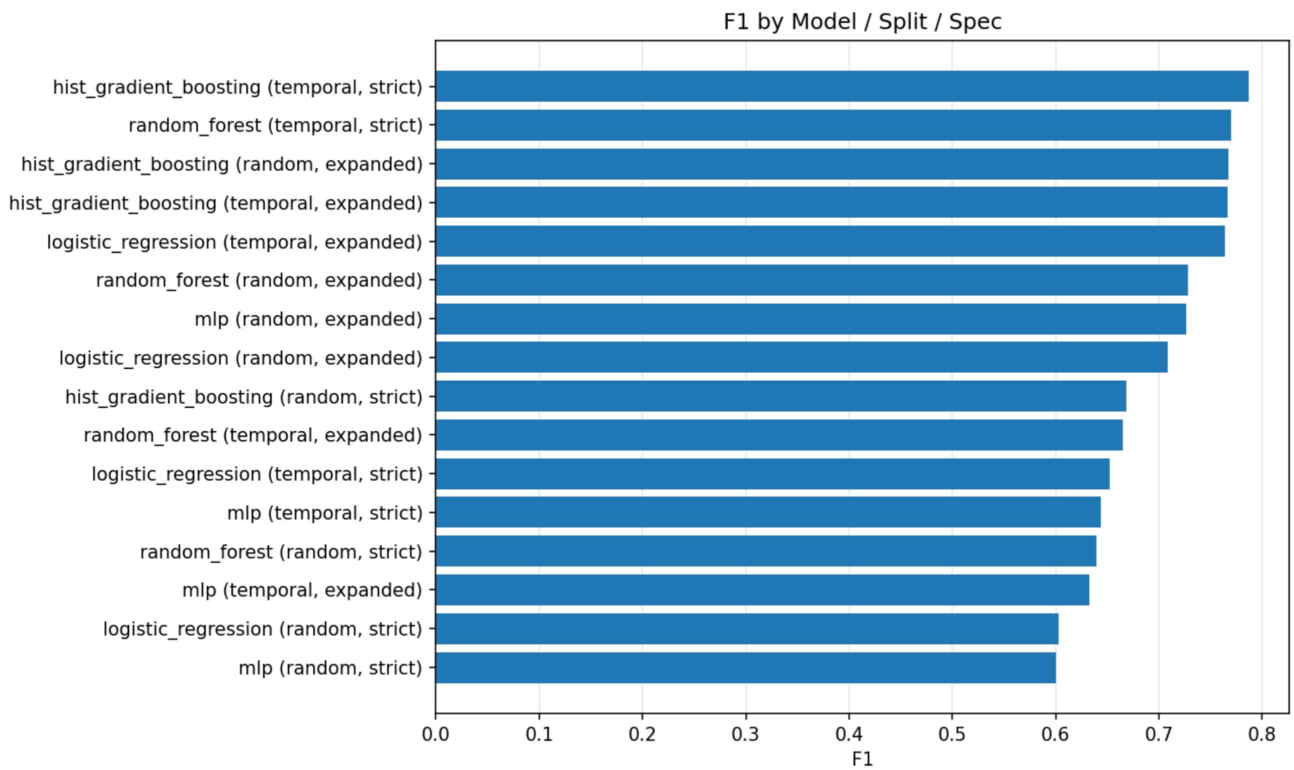

Figure 7. *F1 comparison after validation-based threshold tuning.*

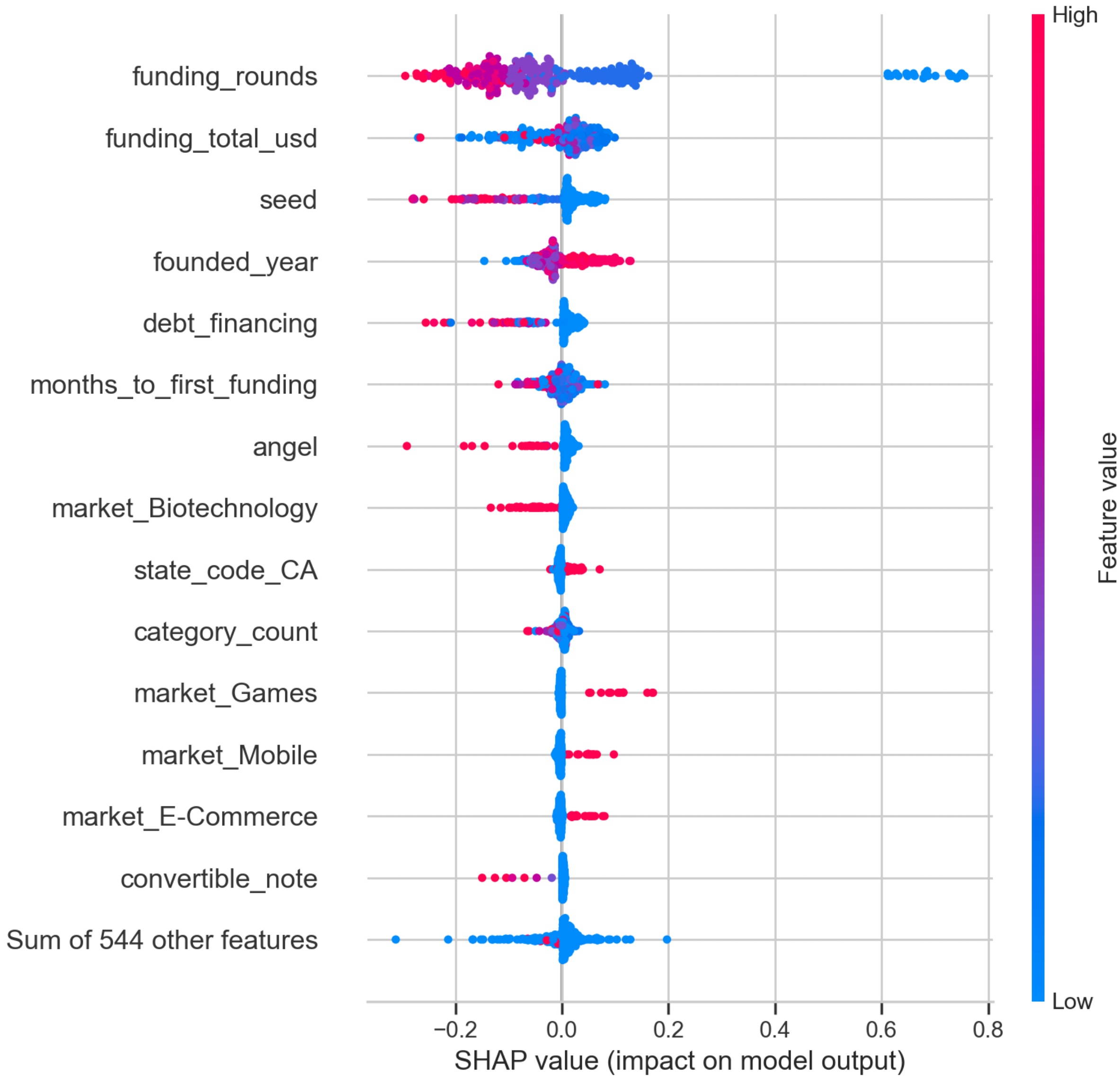

Figure 8. *SHAP beeswarm for the temporal expanded histogram-based gradient boosting model.*

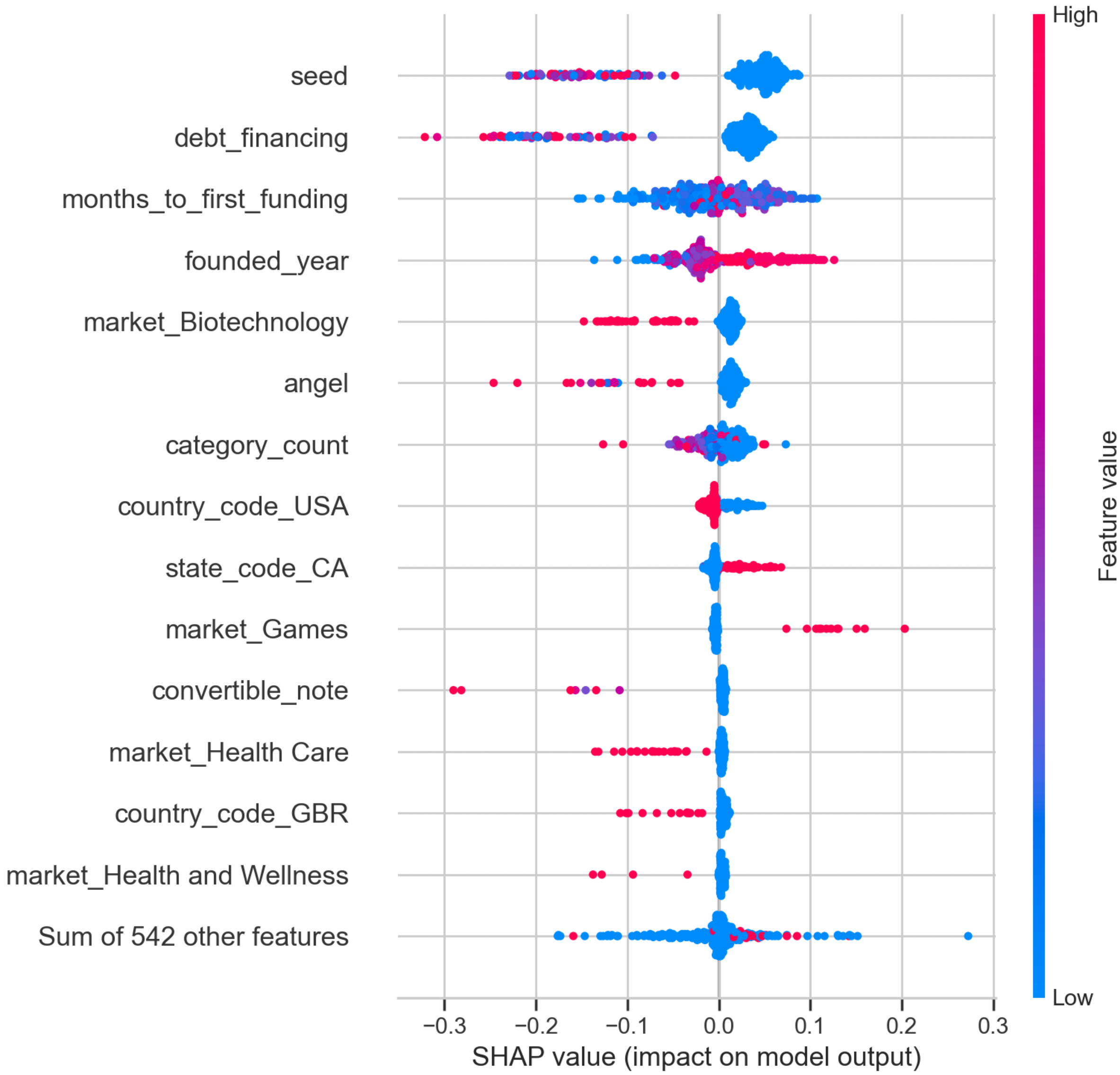

Figure 9. *SHAP beeswarm for the temporal strict histogram-based gradient boosting model.*

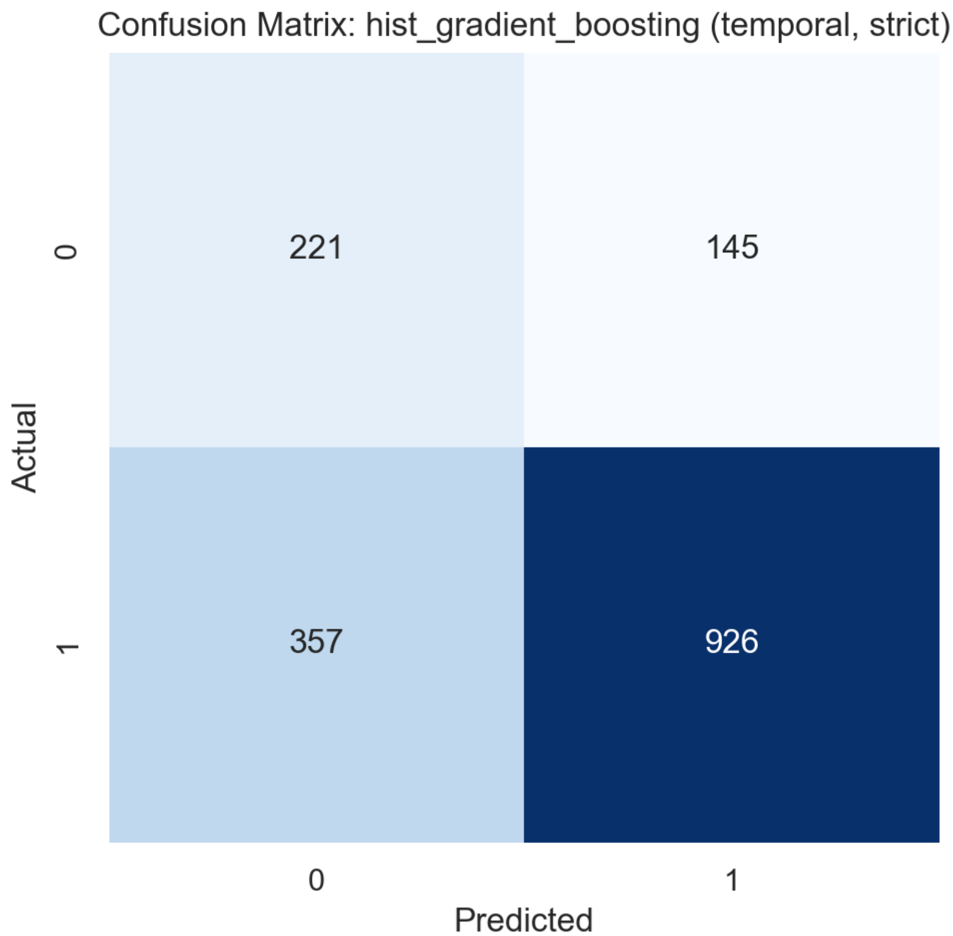

Figure 10. *Confusion matrix for the temporal strict histogram-based gradient boosting model after threshold tuning.*

5. Discussion and Conclusion

The overall picture from these experiments is a mixed but honest one. The models do pick up real signal about which startups are more likely to reach Series B inside a three-year window, but how strong that signal looks depends a lot on how the question is framed. Letting the models see broader funding-history variables pushes performance up noticeably. Closing the information boundary pulls performance back down, though not to the point where the task falls apart. The practical takeaway is that milestone prediction from public startup data is workable, but anyone reporting results from this kind of study should be open about whether they are running a true ex ante screening task or something closer to a snapshot that already carries part of the company's financing story inside it.

The strongest substantive pattern is the importance of financing pace and financing structure. In both the expanded and strict interpretability views, variables tied to funding rounds, seed-stage signals, total capital, and time-to-first-funding play central roles. That finding is consistent with both venture practice and prior research. Early access to capital often proxies for market validation, founder quality, investor networks, and sector attractiveness, even when those underlying mechanisms are not directly observed (Hellmann & Puri, 2002; Ratzinger et al., 2018; Kim et al., 2023; Kim & Lee, 2022; Glücksman, 2020). At the same time, the importance of financing variables is a reminder that the model is not discovering some purely hidden essence of startup quality. It is learning from the market's own prior signals.

A second insight is methodological. This project began as a typical startup-prediction exercise, but the most valuable lessons emerged from evaluation design rather than algorithm shopping. Without a temporal split, it would have been difficult to know whether a strong random-split result reflected real generalization or simply a convenient partition of historical data. Without threshold tuning, some models would have looked artificially poor because they defaulted to majority-class behavior. Without the strict-versus-expanded comparison, it would have been easy to confuse predictive strength with forecasting purity. For a course project, these are important lessons: the design choices around the model can be as consequential as the model itself.

There are also clear limitations. First, the target is approximate because the dataset does not record the exact Series B date. The label should therefore be interpreted as a practical milestone proxy, not a legally or financially exact event-history outcome. Second, the data are historical and partially curated. Public Crunchbase-derived datasets contain missing values, lagged updates, and uneven coverage across markets and periods. Third, the project does not include several variables that could plausibly improve prediction, such as founder biographies, investor reputation, board composition, product text, or patenting activity. Fourth, the project is predictive rather than causal. A strong model does not prove that a feature causes later financing success; it only shows that the feature helps rank firms given the available data.

If the project were extended, the next logical step would be to add unstructured company descriptions or founder text and compare structured-only models with structured-plus-text models, following recent work that incorporates text and sentiment signals into startup-success models (Qiu et al., 2025; Ningrum, 2024). Another useful extension would be a more rigorous event-history design with exact round dates and survival analysis, which would allow the milestone question to be framed in terms of time-to-Series-B rather than simple attainment within a window. A third extension would be robustness testing across geographies and sectors, especially because the present sample is concentrated in a limited set of industries, and recent work has started exploring uncertainty-aware and multi-task formulations that handle this kind of heterogeneity more explicitly (Meng et al., 2026; Weik et al., 2024; Fosfuri et al., 2026).

To wrap up: this case study shows that publicly available startup data can be used to forecast a concrete financing milestone with non-trivial accuracy. Across every experiment I ran, tree-based ensembles were the best-performing model family. Histogram-based gradient boosting was the top overall, random forest stayed close behind and is easier to inspect, and the linear and neural baselines trailed by a visible margin and were less consistent run-to-run. The more important point, though, is about design. Startup prediction is not really a contest between algorithms. It is a question about which information you let the model see, how you handle time when you evaluate it, and how you pick the threshold that turns a probability into a decision. Those are the choices that decide whether a startup-prediction project ends up saying something useful or just looking like it does.

Appendix A. Disclosure of AI Use

This project made limited use of AI tools in ways that the course policy describes as acceptable. I used an AI assistant to brainstorm early framing ideas for the research questions, to help debug scikit-learn and pandas errors while building the training pipeline, and to do grammar, tone, and clarity editing on passages I had already written myself. The research design, the label construction, the choice of feature specifications, the decision to include a temporal split, the threshold-tuning approach, the interpretation of the SHAP and permutation results, and every sentence of the discussion and conclusion are my own work. All numerical results reported in the paper come directly from my own run of the pipeline against the Kaggle Crunchbase dump.

References

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32.

Crunchbase. (2026a). Welcome to Crunchbase data. https://data.crunchbase.com/docs/welcome-to-crunchbase-data

Crunchbase. (2026b). Using the API. https://data.crunchbase.com/docs/using-the-api

Crunchbase. (2026c). 27 top data vendors in 2026. https://about.crunchbase.com/blog/data-vendors

Crunchbase News. (2023). Predicting VC success with Crunchbase data. https://news.crunchbase.com/venture/vc-success-prediction-crunchbase-data-mason-lender/

Corea, F. (2021). Hacking the venture industry: An early-stage startups machine learning ecosystem. Machine Learning with Applications, 6, 100087.

Fosfuri, A., Giarratana, M. S., & Lu, J. W. (2026). Navigating venture capital funding for science-based startups. Research Policy, 55, 105297.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. The Annals of Statistics, 29(5), 1189–1232.

Glücksman, S. (2020). Entrepreneurial experiences from venture capital funding. Venture Capital, 22(4), 305–328.

Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (2nd ed.). Springer.

He, H., & Garcia, E. A. (2009). Learning from imbalanced data. IEEE Transactions on Knowledge and Data Engineering, 21(9), 1263–1284.

Hellmann, T., & Puri, M. (2002). Venture capital and the professionalization of start-up firms: Empirical evidence. The Journal of Finance, 57(1), 169–197.

Kim, D., & Lee, S. Y. (2022). When venture capitalists are attracted by the experienced. Journal of Innovation and Entrepreneurship, 11(1), 31.

Kim, J., Kim, H., & Geum, Y. (2023). How to succeed in the market? Predicting startup success using a machine learning approach. Technological Forecasting and Social Change, 193, 122614.

Kaggle. (n.d.-a). Crunchbase 2013 companies, investors, etc. https://www.kaggle.com/datasets/mauriciocap/crunchbase2013

Kaggle. (n.d.-b). StartUp Investments (Crunchbase). https://www.kaggle.com/datasets/arindam235/startup-investments-crunchbase

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30, 4765–4774.

Mars Discovery District. (2026). Venture capital: Financing a startup in the later stages of growth. https://learn.marsdd.com/article/venture-capital-financing-a-startup-in-the-later-stages-of-growth/

Meng, X., He, Y., & Li, Z. (2026). Uncertainty-aware multi-task learning for startup success prediction and valuation. Ain Shams Engineering Journal, 17, 102006.

Molnar, C. (2022). Interpretable machine learning (2nd ed.). Lulu Press.

Ningrum, I. W. K. (2024). Predicting startup success using machine learning approach. Journal of Applied Informatics and Computing, 8(2), 1–10.

Potanin, M., Chertok, A., Zorin, K., & Shtabtsovsky, C. (2023). Startup success prediction and VC portfolio simulation using CrunchBase data. arXiv. https://arxiv.org/abs/2309.15552

Qiu, Z., Wang, Y., & Li, X. (2025). Enhancing startup financing success prediction based on social media sentiment. Systems, 13(7), 520.

Ratzinger, D., Amess, K., Greenman, A., & Mosey, S. (2018). The impact of digital start-up founders' higher education on reaching equity investment milestones. The Journal of Technology Transfer, 43(3), 760–778.

Razaghzadeh Bidgoli, M., Raeesi Vanani, I., & Goodarzi, M. (2024). Predicting the success of startups using a machine learning approach. Journal of Innovation and Entrepreneurship, 13(1), 80.

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., ... Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography, 40(8), 913–929.

Ross, G., Das, S., Sciro, D., & Raza, H. (2021). CapitalVX: A machine learning model for startup selection and exit opportunity prediction. Journal of Behavioral and Experimental Finance, 30, 100553.

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLoS ONE, 10(3), e0118432.

Schade, P., Tiberius, V., & Bouncken, R. (2023). Predicting entrepreneurial activity using machine learning. Journal of Business Venturing Insights, 19, e00347.

scikit-learn developers. (2026a). LogisticRegression. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

scikit-learn developers. (2026b). RandomForestClassifier. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

scikit-learn developers. (2026c). HistGradientBoostingClassifier. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.HistGradientBoostingClassifier.html

scikit-learn developers. (2026d). MLPClassifier. https://scikit-learn.org/stable/modules/generated/sklearn.neural_network.MLPClassifier.html

Te, Y.-F., Wieland, M., Frey, M., Pyatigorskaya, A., Schiffer, P., & Grabner, H. (2023). Making it into a successful Series A funding: An analysis of Crunchbase and LinkedIn data. The Journal of Finance and Data Science, 9, 100099.

Wei, C.-P., Chen, Y.-L., & Huang, J.-S. (2025). To shine or not to shine: Startup success prediction by combining technological and venture-capital-related features. European Journal of Operational Research, 323(2), 602–618.

Weik, S., Fesenfeld, L., & Eberle, D. (2024). Venture capital and the international relocation of startups. Research Policy, 53(9), 105076.

Żbikowski, K., & Antosiuk, P. (2021). A machine learning, bias-free approach for predicting business success using Crunchbase data. Information Processing & Management, 58(4), 102555.

Zhang, C., Zhang, H., & Hu, X. (2019). A contrastive study of machine learning on funding evaluation prediction. IEEE Access, 7, 106307–106315.